The Hehl–Datta Coefficient from Relative-Entropy Positivity (Final Updated Version): Sign, Rigidity, and Pure-Information Magnitude

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August 14, 2026 — Final Updated Version

Abstract

This is the final updated version of Paper V of the Emergent Gravity Program. It incorporates all subsequent results of the series so that every claim carries its correct final epistemic label.

Sign [**P**]: relative-entropy positivity forces the contact term negative. **Magnitude by rigidity** [**P**]: given the Ryu–Takayanagi normalization and the modular selection rule, the coefficient is uniquely $-3\kappa/16$. **Magnitude from pure information** [**P**]: the Fisher–canonical matching is closed by the null-plane limit together with the absorbability of the finite-ball residue.

The multi-digit numerical coefficient of the kinematic residue $\mathcal{I}_B$, the de Sitter domain, the non-linear regime, and UV completion remain [**O**]. No outdated open labels appear on claims that have been closed.

Why this update is required. The original Paper V left the pure-information magnitude open because it depended on the then-unevaluated integral $\mathcal{I}_B$. That integral has been shown to be non-vanishing and absorbable, and the matching has been closed. This version brings Paper V into full consistency with the rest of the series.

Epistemic Conventions

[**P**] proved/derived; [**C**] conjectured with support; [**O**] open. Numerical agreement is not proof. Consensus is not proof.

1 Final Status of the Claims of This Paper

1. Sign of the Hehl–Datta coefficient (relative-entropy positivity). [**P**]
2. Magnitude by rigidity (RT normalization + modular selection rule). [**P**]
3. Magnitude from pure information (null-plane limit + absorbability). [**P**]
4. Resulting coefficient $\mathcal{L}_{\mathrm{HD}} = -3\kappa/16\, \mathcal{J}_5 \cdot \mathcal{J}_5$. [**P**]
5. Multi-digit numerical coefficient of the residue $\mathcal{I}_B$. [**O**]

6. de Sitter / cosmological domain. **[O]**
7. Non-linear regime and UV completion. **[O]**

2 Cross-References to the Rest of the Series

This paper is consistent with and cites:

- Paper IV (Final Updated) — unconditional metric sector, non-vanishing and absorbable residue
- Paper VI — evaluation establishing non-vanishing of $\mathcal{I}_B$
- Paper VII — explicit closure of the Fisher–canonical matching
- Paper VIII (Version 4) — order-of-magnitude coefficient ($\mathcal{O}(1)$) and remaining analytic requirement
- Paper IX — domain of validity (AdS closed, de Sitter open)
- Final Status document (14 August 2026) — authoritative label table for the whole program

3 Summary

The Hehl–Datta coefficient is now determined in sign and in magnitude by quantum-information principles inside the AdS regime. The only open items that remain are those that lie outside the reach of the present methods. All labels in this version reflect that final accounting.

References

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